

Mean, Median, and Mode

Lesson 8-2

Name: _____

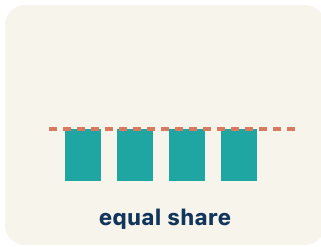
Date: _____

Class: _____

Key Vocabulary

Level 1 support

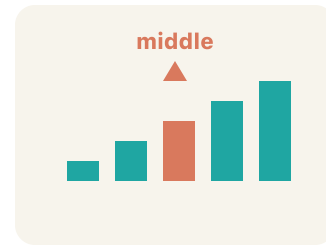
Picture first, then the word, then a plain-language meaning. Say each word out loud.



Mean of 2, 4, 6: add them ($2+4+6=12$), divide by 3
→ mean = 4

Mean

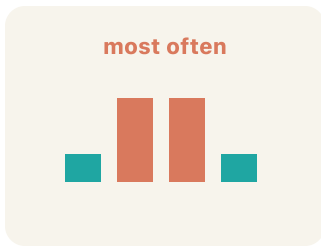
The average. Add all the numbers, then divide by how many there are.



Data: 1, 3, 5, 7, 9 → the middle (3rd) value is 5, so
median = 5

Median

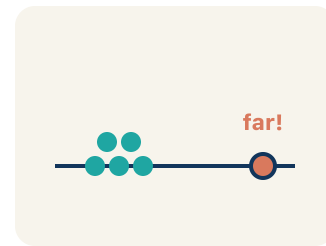
The middle number when you put them in order.



Data: 2, 3, 3, 5, 7 → 3 appears twice (most often),
so mode = 3

Mode

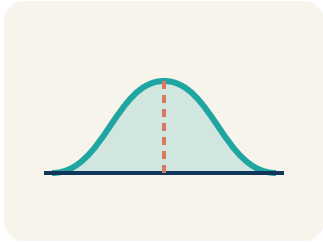
The number that shows up most often.



Data: 10, 12, 11, 13, 50 → 50 is far from the others,
so 50 is an outlier

Outlier

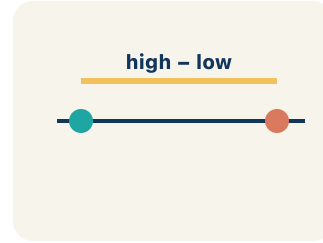
A number that is much bigger or smaller than the rest.



If most values cluster in the center with fewer at the ends, the distribution is symmetric

Data distribution

How the data looks: where it sits and how spread out it is.



8, 9, 10, 11 has low variability (close together); 2, 10, 25, 40 has high variability (far apart)

Variability

How spread out the numbers are.

Key Ideas & Notes



WHAT WE'RE LEARNING TODAY

I can find the mean, median, and mode of a data set.

1 Notice

You're the stats analyst for the school basketball team.

2 Key idea

I can find the mean, median, and mode of a data set.

3 Words I'll use

Mean

Median

Mode

Outlier

Data distribution

Variability



Think About It

- What are the highest and lowest scores?
- Do any scores repeat?
- Which score seems like a typical game for this team?

See the notes in action: watch one worked all the way through, then try the next with the same steps.

I do — watch

Follow each step as your teacher solves it.

Problem: Find the mean of: 10, 14, 8, 12, 16

- A. 12
- B. 14
- C. 10
- D. 60

Step 1 Sum = $10 + 14 + 8 + 12 + 16 = 60$.

Step 2 Mean = $60 \div 5 = 12$.

Answer: A. 12

 We do — together

Solve this one with your class using the same steps.

Problem: Find the median of: 3, 7, 2, 9, 5

- A. 5
- B. 3
- C. 7
- D. 5.2

Step 1

Step 2

Answer:

 You do — your turn

Now try one on your own. Show every step.

Problem: Find the mode of: 4, 7, 4, 9, 2, 4, 8

- A. 4
- B. 7
- C. 9
- D. 2

Show your work:

Turn & Talk — Discussion Points

Level 1 support

Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

The team scored 42, 38, 55, 41, 38, 44, 40 points over 7 games. Which 'average' would best tell the coach the team's typical scoring: mean, median, or mode?

Level 1 support

Start here: Mean, median, and mode each describe the center of the data in a different way.

 **Need a hint?**

Sentence starters (optional)

I would use the ____ because it best shows the typical score.

Usaría la ____ porque muestra mejor el puntaje típico.

The ____ is the most common score, which is ____.

La ____ es el puntaje más común, que es ____.

Word bank: mean median mode average data set

Listen for: Students distinguish the three measures and justify a choice, noting the mode (38) is the most repeated and the mean/median sit near the center.

Level 2

Justify: which measure would change the most if one game's 55 were actually 100? Explain why.

- The ____ would change most because it uses ____.
- The mode and median would stay the same because ____.

EXPLORE

After ordering the scores on the number line, how do you find the median, and how is it different from the mean you calculated?

Level 1 support

Start here: The median is the middle of ordered data; the mean is the equal-share total.

 Need a hint?

Sentence starters (optional)

To find the median, I first ____ the scores, then pick the ____.

Para hallar la mediana, primero ____ los puntajes y luego elijo el ____.

The mean is different because I have to ____ and ____.

La media es diferente porque tengo que ____ y ____.

Word bank: mean median mode order data set

Listen for: A strong answer orders the data, finds the middle value as the median, and explains the mean is found by summing all values and dividing.

Level 2

Compute and compare: find the mean and median of 42, 38, 55, 41, 38, 44, 40. Are they close? What does that tell you?

- The mean is ____ and the median is ____.
- They are close/far apart, which tells me the data is ____.

CONNECT

Quiz scores were 85, 92, 78, 88, 92, 95, 72, 88, 92, 84. The teacher says 'most scored around 87'; a student says 'the most common score was 92.' Who used the mean, and who used the mode?

Level 1 support

Start here: Different people pick different measures of center to make their point.

 **Need a hint?**

Sentence starters (optional)

The teacher used the _____, and the student used the _____.

El maestro usó la _____, y el estudiante usó la _____.

The _____ is more useful here because _____.

La _____ es más útil aquí porque _____.

Word bank: mean median mode average data set

Listen for: Students identify 87 as the mean and 92 as the mode, and argue which is more useful for describing the class's typical performance.

Level 2

Critique: is the student technically right that 92 was 'the most common,' yet possibly misleading about how the class did? Defend.

- The student is right that the mode is _____, but it can mislead because _____.
- A fairer summary of the whole class is the _____ since _____.

REFLECT

For the data set 4, 7, 10, 7, 12, talk through how you find the median.

Level 1 support

Start here: Order the values, then find the middle one.



Need a hint?

Sentence starters (optional)

First I order the values: ____.

Primero ordeno los valores: ____.

The median is ____ **because it is in the** ____.

La mediana es ____ *porque está en el* ____.

Word bank: median mean mode order middle

Listen for: Students order to 4, 7, 7, 10, 12 and identify 7 as the median (middle value).

Level 2

Generalize: explain how to find the median when a data set has an EVEN number of values.

- With an even number of values, I ____ the two middle numbers.
- For example, with 4, 6, 8, 10 the median is ____ because ____.

Write About the Math

The Writing Revolution

I can explain my reasoning using the words mean, median, mode, and outlier.

1. Kernel Sentence subject + verb

Model: Mean is the average. Add all the numbers, then divide by how many there are.

Media es el promedio. Suma todos los números y divide entre cuántos hay.

Write a kernel sentence about mean. Use a subject and a verb.

Escribe una oración base sobre media. Usa un sujeto y un verbo.

2. Sentence Expansion because · but · so

Kernel: Mean matters in math

Media importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because
porque

Mean matters in math because ____.

Media importa en matemáticas porque ____.

but
pero

Mean matters in math, but ____.

Media importa en matemáticas, pero ____.

so
entonces

Mean matters in math, so ____.

Media importa en matemáticas, entonces ____.

3. Sentence Types 4 ways to write a math idea

Statement Afirmación

Tell one true fact about mean.
Di un hecho verdadero sobre mean.

Mean ____.

Question Pregunta

Ask a question about mean.
Haz una pregunta sobre mean.

How does ____ ?

¿Cómo ____ ?

Exclamation Exclamación

Show excitement about mean.
Muestra entusiasmo sobre mean.

Wow, ____ !

¡Guau, ____ !

Command Mandato

Tell a partner what to do with mean.
Dile a un compañero qué hacer con mean.

First, ____ .

Primero, ____ .

4. Explain Your Reasoning use a sentence starter

I would use the ____ **because it best shows the typical score.**

Usaría la ____ *porque muestra mejor el puntaje típico.*

The ____ **is the most common score, which is** ____.

La ____ *es el puntaje más común, que es* ____.

To find the median, I first ____ **the scores, then pick the** ____.

Para hallar la mediana, primero ____ *los puntajes y luego elijo el* ____.

| Try It

Solve on your own. Check the answer key when you are done.

1. What is the mode of 2, 5, 5, 7, 9, 5?

- A. 5
- B. 7
- C. 2
- D. 9

Show your work:

2. What is the median of 4, 8, 10, 14?

- A. 9
- B. 10
- C. 8
- D. 12

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A basketball player scored 15, 18, 16, 42, and 14 points in 5 games. Find the mean and median. Which better represents the player's typical game? Explain how the outlier affects each measure.

Sentence starter: Mean = _____. Median = _____. The _____ better represents a typical game because the outlier (_____) pulls the _____ toward it, making it _____ than most scores. The median is not affected because _____.

Show your work:

Reflect — Exit Ticket

Data set: 4, 7, 10, 7, 12. What is the median?

- A. 7
- B. 8
- C. 10
- D. 4

Your answer:

Answer Key & Teacher Guide

1. **We Try (notes):** A. 5 — *Ordered: 2, 3, 5, 7, 9. The middle value is 5.*
2. **You Try (notes):** A. 4 — *4 appears three times — more than any other value. Mode = 4.*
3. **Try It 1:** A. 5 — *5 appears most often (three times).*
4. **Try It 2:** A. 9 — *With four values, average the middle two: $(8 + 10) \div 2 = 9$.*
5. **Exit Ticket:** A. 7 — *Ordered: 4, 7, 7, 10, 12. The middle (3rd) value is 7.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Mean is the average. Add all the numbers, then divide by how many there are.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).